

Problem

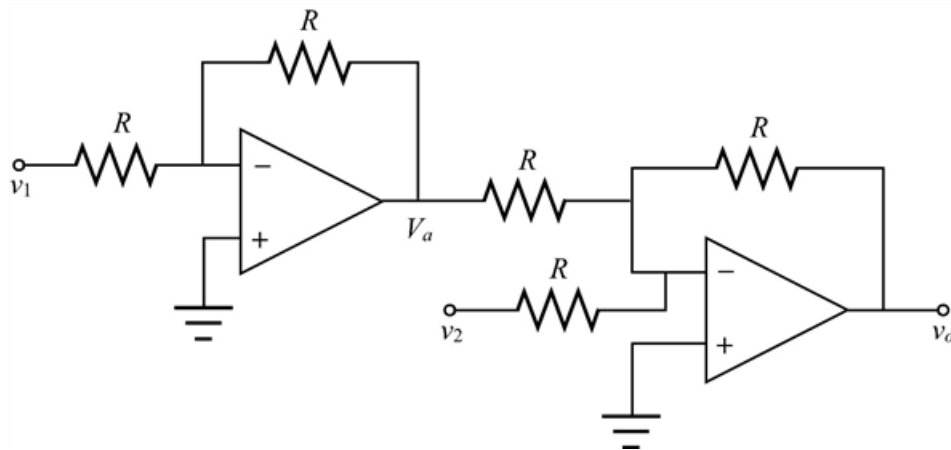
Using two op amps, design a subtractor.

Step-by-step solution

Step 1 of 3

The subtractor circuit is achieved by cascading an inverting amplifier and two input inverting summer.

Consider the following subtractor circuit.



Step 2 of 3

From the subtractor circuit, the non-inverting terminals of both op-amps are connected to ground. So, the voltages at non-inverting terminals of both op-amps are zero.

According to virtual ground concept, the voltages at inverting and non-inverting of the op-amp are same for ideal op-amp.

Therefore, the voltages at inverting and non-inverting of the both op-amps are zero.

$$\begin{aligned} v_- &= v_+ \\ &= 0 \text{ V} \end{aligned}$$

Apply Kirchhoff's current law at inverting terminal of the left side op-amp.

$$\begin{aligned} \frac{v_- - v_1}{R} + \frac{v_- - V_a}{R} &= 0 \\ \frac{0 - v_1}{R} + \frac{0 - V_a}{R} &= 0 \\ \frac{-v_1 - V_a}{R} &= 0 \\ V_a &= -v_1 \end{aligned}$$

Step 3 of 3

Apply Kirchhoff's current law at inverting terminal of the right-side op-amp.

$$\frac{v_- - v_2}{R} + \frac{v_- - V_a}{R} + \frac{v_- - v_o}{R} = 0$$

$$\frac{0 - v_2}{R} + \frac{0 - V_a}{R} + \frac{0 - v_o}{R} = 0$$

$$\frac{-v_2 - V_a - v_o}{R} = 0$$

Substitute $-v_1$ for V_a .

$$-v_2 - (-v_1) - v_o = 0$$

$$v_o = v_1 - v_2$$

The output is the difference of voltages v_1 and v_2 .

Hence, the subtractor circuit using two op-amps is designed.